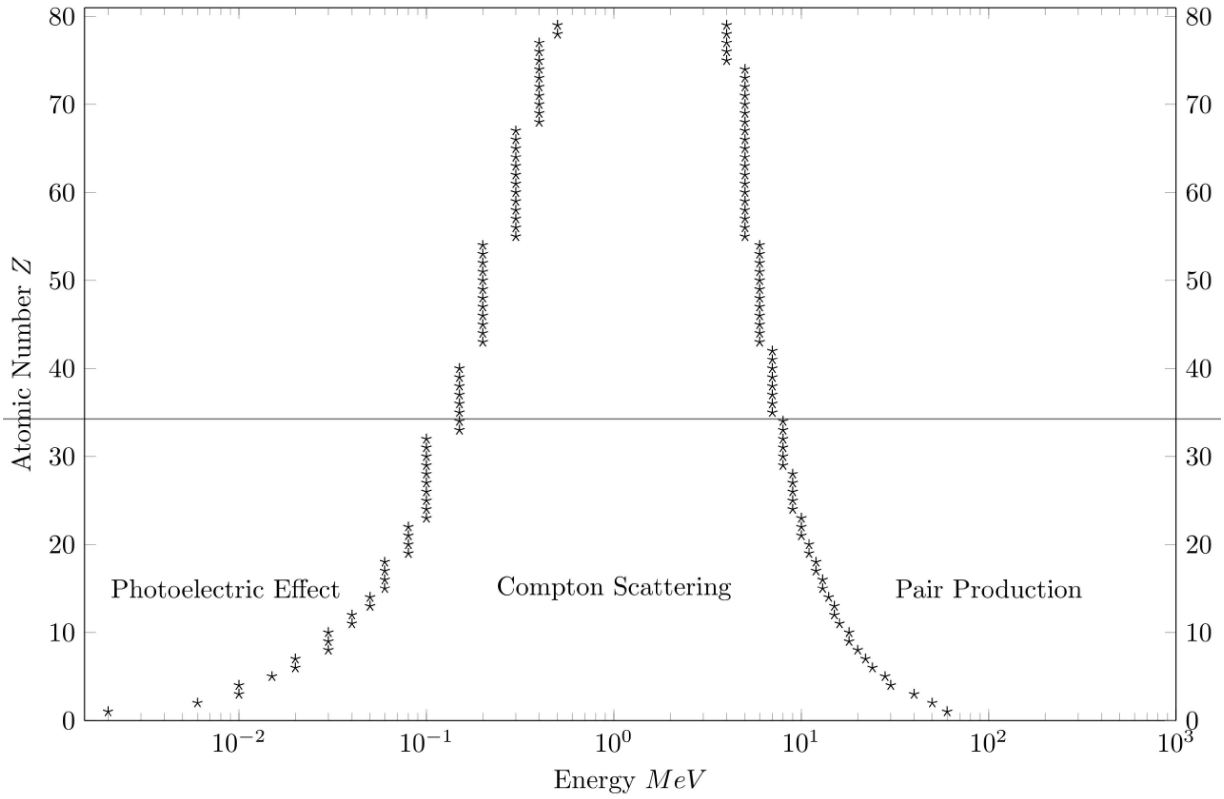
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File:Dominant Photon-Matter Interaction.svg



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Original file (SVG file, nominally 608 × 401 pixels, file size: 134 KB)

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Captions Edit

English Add a one-line explanation of what this file represents

Summary

Description	**English:** Plot showing the relative importance of the photoelectric effect, Compton scattering and pair production with different Z and mean photon energy. Data taken from the NIST XCOM database (https://dx.doi.org/10.18434/T48G6X). For any given element (atomic number Z) a star is added at the energy where the cross section value for the process on the right becomes larger than the cross section value for the process on the

	left. For calcium (Z=20), Compton scattering starts to dominate at *hu*=0.08 MeV and ceases at 12 MeV.
Date	26 May 2018
Source	Own work
Author	Michael Weinold

Graph plotted with pgfplot. Feynman Diagrams plotted with tikz-feynman.

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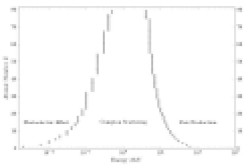
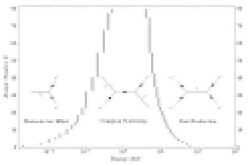
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File history

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	Date/Time	Thumbnail	Dimensions	User	Comment
current	**07:13, 24 June 2024**		608 × 401 (134 KB)	Michael Weinold (talk | contribs)	Removed Feynman Diagrams (some of which were not depicting the correct interactions)
	11:26, 26 May 2018		570 × 376 (175 KB)	Michael Weinold (talk | contribs)	User created page with UploadWizard

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File usage on other wikis

The following other wikis use this file:

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